# Ramsha Khan

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## Technologies

**Languages:** Python, C++
**Framework/Libraries:** PyTorch, Tensorflow, Hugging Face, scikit-learn, Docker, Git, GitHub, Linux
**Technical Skills:** Exploratory Data Analysis, Natural Language Processing, Machine Learning, Deep Learning, LLMs, CI/CD

## Experience

**Research Intern at FrontiersMind** - Bengaluru, Karnataka — Aug 2026 – Oct 2026

- Built **Triton decode kernels** for Grouped Value Attention (fused query and value-only cache append, sequence-split flash-decoding with key-map absorption) and **CUDA-Graph decoding**, cutting KV-cache memory **48%** and making attention **2.5×** faster and end-to-end decoding **1.6–1.8×** faster than **flash-attn GQA** at up to 32k context on an RTX 3090.
- Contributed upstream PRs adding Lumma-0.6B-Base architecture support to Hugging Face Transformers and llama.cpp, including GGUF conversion and llama.cpp inference integration.
- Built **pretraining** data pipelines for 10 Indic languages (deduplication, quality filtering, transliteration) to improve corpus consistency and coverage.
- Developed **evaluation and benchmarking pipelines** spanning pretraining and post-training, enabling systematic comparison of model checkpoints, variants, and task-level performance.
- Pretrained and benchmarked **Lumma-Fev (Decision model)** variants across structured decision tasks, conducting qualitative and behavioral evaluations to analyze failure modes and performance differences.

**AI Research Intern at AI Institute of South Carolina (AIISC)** - Remote — April 2026 – June 2026

- Built a **synthetic preference dataset pipeline** to curate high-quality dataset for LLM post-training and implemented post-training alignment techniques including **LoRA-based Supervised Fine-Tuning (SFT)** and **Direct Preference Optimization (DPO)** for domain-specific Small Language Models (SLMs), reducing DPO loss by **77%** and improving reward margin by **3.5x**.
- Conducted qualitative evaluation comparing SFT and DPO model outputs across diverse customer-support intents, identifying training signal gaps to guide preference dataset iteration.

**Machine Learning Engineering Intern at Matrice AI** - Remote — March 2026 – May 2026

- Fine-tuned YOLO-based object detection models for conveyor belt footage using domain-specific augmentations, improving accuracy by **12%** and reducing false positives by **20%**.
- Built real-time video analytics pipelines processing 20–25 FPS streams for object tracking, alerting, and monitoring.
- Extended the BYOM platform with end-to-end support for RF-DETR and RT-DETR, covering **training, evaluation, export, and inference**; benchmarked MOT algorithms, improving tracking stability by **10%** under occlusion and motion blur.

## Projects

**FinetuneX** — github FinetuneX

- Built a modular framework from scratch to **fine-tune LLMs** supporting **multiple self-implemented LLM architectures** with reusable architectural components designed for easy extension to new LLM architectures.
- Wrote custom **Triton kernels for Flash Attention v2 supporting Grouped Query Attention**, achieving **15-30x** speedup and **10-20x** memory reduction benchmarked against standard pytorch.
- Implemented Supervised Fine-Tuning (**SFT**), Direct Preference Optimization (**DPO**) and Implicit Preference Optimization (**IPO**) pipelines to significantly enhance post-training model alignment and behavioral accuracy.
- Integrated support for **distributed training strategies including Distributed Data Parallel (DDP), Fully-Sharded Data parallelism (FSDP)**, and memory-efficient training techniques.

**LexGround AI** — github LexGround

- Built an AI-powered legal document intelligence system using **FastAPI** that processes PDFs, scanned images, and DOCX files, leveraging **Docling-based OCR** and document parsing to extract structured information and generate case summaries.
- Implemented **semantic retrieval pipeline** using Sentence Transformers and FAISS, Dockerized deployment workflows, and feedback-driven learning mechanisms for document understanding, entity extraction, and automated draft generation.

**Intracranial Aneurysm Detection** — github Aneurysm Detection

- Built a **multi-modal 3D imaging pipeline** (CTA, MRA, T1/T2 MRI) to automate early detection of brain aneurysms; leveraged **RSNA 2025 Kaggle dataset** and trained **EfficientNet** on **4,348** brain scans.
- Preprocessed 3D DICOMs brain scans as 2D multi-channel tensors, cutting per-epoch time by **60x**; optimized training with **Weighted Binary Cross-Entropy**, achieved validation loss of **0.25** on multi-label classification.

**Retinal Vessel Segmentation** — github Segmentation

- Trained ResUNet-Deeper for retinal vessel segmentation, achieving **0.887 Dice** with 12 augmentations and cross-dataset evaluation; optimized the model with **ONNX + quantization**, reducing model size by **50.2%** and doubling inference speed.

## Education

**University of Mumbai**, B.E in Computer Engineering CGPA: 8.5/10.0 — July 2023 – May 2026
**Technical Diploma** in Computer Engineering CGPA: 9.1/10.0 — Aug 2020 – June 2023